Protocol for a Meta-Research Study

A Review of Risk of Bias and Trustworthiness Profiles of Retracted Trials in Cochrane Reviews

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Introduction

Randomised controlled trials (RCTs) are widely regarded as the most reliable method for evaluating the effectiveness of a treatment or intervention. Moreover, systematic reviews aim to synthesise data from multiple studies that meet pre-specified eligibility criteria to answer a specific research question, possibly using meta-analysis. In view of this, systematic reviews of RCTs enable one to make the most appropriate healthcare decisions by synthesising all the empirical evidence from the eligible RCTs in the literature; however, the outcome depends heavily on the quality of the RCTs it contains, both in terms of risk of bias and trustworthiness, because studies with high risk of bias or serious trustworthiness concerns may significantly affect the evidence presented in the analysis.

Commented [JW1]: looking good - might suggest some minor changes to wording to improve readability/ flow, but this is looking good

Bias or systematic error is defined as a consistent deviation from the true value that impacts the accuracy of results[1]. In RCTs, several potential sources of bias can arise, for example, in how participant outcomes are measured or in design flaws, such as a lack of allocation sequence concealment. However, it is usually impossible to know to what extent biases have affected the results of a particular study or analysis[2]. We define trustworthiness separately from bias risk; an RCT is untrustworthy or problematic if there are serious questions regarding the trustworthiness of its data or findings, which leads to losing confidence in the findings or conclusions [3]. Causes of trustworthiness range from unintentionally poor research practices, such as inadequate study design and data collection methods, to deliberate scientific misconduct, including data fabrication and plagiarism. Although both bias and trustworthiness issues can impact the results of a study, trustworthiness relates to the overall credibility of the study.

Commented [GL2]: From Lisa remark:

I always struggle with the definition... defining trustworthiness as trustworthiness! Could add that this means losing confidence in the findings / conclusions?

Commented [JW3]: This is always tricky - perhaps say something like "trustworthiness assessment involves checking for serious research integrity issues which may be caused by researcher misconduct or honest but critical errors in the delivery of the study"

Commented [JW4]: Is there some repetition between this and the earlier text "we define trustworthiness..." to end of that paragra[ph?

Systematic reviews are commonly used to guide clinical decisions, and the inclusion of biased results within them is likely to lead to incorrect conclusions and sub-optimal clinical decisions. For this reason, to evaluate the likelihood that systematic errors of the study design or conduct will give misleading results, RCTs included in systematic reviews are routinely appraised using risk-of-bias tools, such as RoB1 and RoB2, endorsed by Cochrane, with RoB2 being the latest version[4]. Consequently, the Cochrane Handbook for systematic reviews suggests that risk-of-bias assessments should be used to determine eligibility for primary analyses or to stratify trials according to bias risk [1]. The INSPECT-SR (INveStigating ProblEmatic Clinical Trials in Systematic Reviews) tool has been developed to assess the trustworthiness of RCTs using their trial reports and associated documentation[3]. INSPECT-SR has also been endorsed by Cochrane and would be applied before the risk of bias assessment and after screening for potentially included studies. INSPECT-SR guidance recommends that trustworthiness concerns should be clearly reported, and studies with serious concerns should be excluded from a systematic review, as a problematic study provides unreliable information[3].

Commented [GL5]: From Lisa remark:

Need to say what RISK of bias is .. Not everyone understands this

Commented [JW6]: Cochrane endorses both RoB1 and 2 - 2 is in the latest version of the Handbook

Commented [SR7]: Systematic reviews have not been mentioned before this point. The idea of systematic reviews of trials needs to be introduced and the relevance explained I think

Commented [JW8R7]: Think it is mentioned above ("systematic reviews look to combine data..."

Commented [SR9R7]: That text is mine ... I think Wilfried has accepted my suggested text already maybe

Commented [LB10]: I think it is important for people to understand where this falls in the SR process

A retracted study exemplifies a situation where serious trustworthiness concerns have been raised. We recall that a retracted study is a published scientific paper that has been invalidated or withdrawn from the academic record by the journal's editor or the study's author. The most prevalent reasons for retraction include fraudulent peer reviews and unreliable data[5]. As a tool for assessing trustworthiness, studies flagged as problematic by INSPECT-SR may subsequently be retracted. We will explore the connection between INSPECT-SR and retractions.

Commented [GL11]: Added because of Lisa comment:

I would introduce retractions in a separate para, and potentially start with a definition of retracted studies, as you did for ROB and trustworthiness - also establish the link to INSPECT-SR: INSPECT may identify studies eligible for retraction, or studies identified as problematic by INSPECT may later be retracted. We are going to examine the link between INSPECT SR and retractions.

A retracted article should not be included in a systematic review that informs healthcare decision-making. However, it can often take years for a retraction to occur, and many are never retracted[6], during which time the problematic study continues to affect healthcare decisions through the systematic review. This suggests that systematic reviewers cannot rely on retraction status alone to identify problematic studies[7]. It is therefore pertinent to consider the INSPECT-SR and RoB profiles of these studies and to ask if these tools would identify them ahead of their being retracted. It is important to establish whether, if the associated guidance for these tools were followed, trials' influence could be prevented, thereby ensuring that systematic reviews are based on the most reliable evidence available.

Moreover, it would also be useful to know more about the relationship between INSPECT-SR and RoB assessments, the motivation being that, if these studies are all high risk of bias and the handbook guidance is followed, trustworthiness assessment might be redundant for the purpose of safeguarding systematic reviews. During Stage 2 of the INSPECT-SR project, provisional trustworthiness checks indicated that some systematic reviews included trials with substantial trustworthiness concerns that were not clearly captured by risk-of-bias judgements, but these findings were exploratory and predated the development of the INSPECT-SR tool[8].

1. Objectives

Primary objective:
To evaluate whether INSPECT-SR would identify studies retracted for trustworthiness reasons.

Secondary objective:
To examine the relationship between risk-of-bias and trustworthiness judgments in retracted RCTs using RoB1 and INSPECT-SR.

Commented [GL12]: From Neil comment

2. Methodology

This is a methodological study that employs two assessment tools: the "Cochrane Risk of Bias tool" (RoB)[1] and INSPECT-SR, applied to a sample of retracted randomised controlled trials (RCTs) within systematic reviews. We will select retracted trials that have already been evaluated for risk of bias in published Cochrane systematic reviews. The rationale for selecting Cochrane systematic reviews is to ensure that the risk of bias has already been assessed using the RoB1 tool endorsed by Cochrane. Additionally, concentrating on retracted trials will allow us to determine whether INSPECT-SR would have identified issues in these studies before their retraction.

2.1. Eligibility Criteria

We will select retracted RCTs from reviews in which risk of bias has been assessed using the earlier Cochrane-endorsed tool (RoB1), as this remains the most commonly applied tool in existing Cochrane systematic reviews[9]. RoB2 will not be included, as its more recent adoption would likely yield only a small number of eligible studies, limiting the potential for meaningful analysis. Additionally, since the RoB1 is sometimes evaluated at the outcome level, if multiple assessments exist, we will consider only the first risk of bias assessment outcome available in the systematic review report.

Commented [GU13]: I think retracted whatever the reason, in answer to Neil.

Commented [SR14]: I think this is a different issue so 'furthermore' does not really work here. Similarly for 'additionally'.

2.2. Trustworthiness Assessment Procedure

Given the potential challenges in assessing the trustworthiness of studies known to be retracted and how this may influence an assessor's judgement, the trustworthiness judgement will be transparently justified and reported using the INSPECT-SR framework by two independent assessors. The assessors will first discuss any disagreements, and if necessary, they will consult a third reviewer.

The INSPECT-SR tool comprises a series of questions divided into four domains to guide the reviewer in conducting a trustworthiness assessment. Each domain is assessed separately, and an overall judgement will subsequently be provided by the assessor[3]. The first domain includes a check for post-publication retraction notices; however, this question will be deemed redundant in our context, as we are exclusively examining retracted RCTs. Therefore, this question will be omitted from the assessment of this domain. For the second and third checks i.e. checks 1.2 and 1.3 (checks of post-publication notices and track record of the author team), we will consider the dates of any notices in relation to the publication date of the Cochrane Review, and will assess them using only notices that would have been available at that date (such that expressions of concern published after the Cochrane Review would not be included in the assessment). The assessor will also extract the following details about each trial: the topic area (drugs, vaccines, trials on medical devices, therapy trials, trials on nutrition, surgery, complementary medicine and others), study design (individual randomised or cluster randomised trials), description of the intervention, number of people randomised, outcome utilised for RoB1 assessment, type of outcome assessed (subjective or objective), and year of publication.

2.3. Search Strategy

To identify eligible studies, we will utilise the Feet of Clay Detector available on the Problematic Paper Screener website, developed by Guillaume Cabanac, which facilitates the quick identification of publications with "annulled" references (i.e., retracted, removed, or withdrawn) in their reference lists[10].

We will identify Cochrane systematic reviews that include "annulled" references by selecting "Cochrane Database of Systematic Reviews" as the "Venue" in the Feet of Clay Detector Database. Following this, we will review these references to identify which ones are actual trials included in the studies list of the Cochrane Review (as opposed to being mentioned in the background or discussion, for example) with an available RoB1 assessment. To ensure a sample of trials from recently published reviews, we will sort the systematic reviews by publication year (most recent to oldest) and work down the list. If a review contains more than one retracted trial, we will select only one study using the following principle: Annulled studies are organised in a random order within the Feet of Clay Detector database; we will evaluate these studies for eligibility in the same order and select the first eligible study identified. We will cease selection once we achieve our target sample size. Importantly, a study can only be selected

Commented [JW15]: or maybe just name the checks 1.2 and 1.3 here?

Commented [GL16]: I am still thinking about this one. I fail to find an ideal literature that will give us an overall classification that will suit our sample size; some are too detailed. I am thinking about going with the 7 categories given here: https://pmc.ncbi.nlm.nih.gov/articles/PMC2768680/

Commented [SR17R16]: Which 7 categories are you referring to? The set of 7 that I can find aren't all relevant to trials.

Commented [GL18R16]: I am also not sure that all are relevant, but I was thinking about: trials on medicinal products (drugs, vaccines), trials on medical devices, therapeutic trials, Diagnostic studies, trials on nutrition, and others.

Commented [SR19R16]: Diagnostic studies generally not trials (you can have a trial of a diagnostic tool but I don't think that is what this is referring to). Medicinal products/medical devices/therapeutic interventions/behaviour change (or complex interventions)/other perhaps

Commented [JW20R16]: surgery, complementary medicine might be worth identifying

Commented [JW21]: perhaps cite it and attribute it to Guillaume that way?

Commented [JW22]: does this need an explanation?

Commented [JW23]: does this need an explanation?

Commented [JW24]: do you need to say e.g. "We will do this by navigating to the review on the Cochrane library and manually locating the trial in the reference list?" - it may be in included studies, or somewhere else

Commented [SR25]: How?

Commented [SR26]: What determines this order? It seems important to know.

Commented [JW27R26]: I think based on our previous conversations it is by date they were added to the Feet of Clay database?

once; this means that if we choose a trial from a particular systematic review, that trial will not be eligible for selection from any other reviews.

The extracted data at the end of this process will consist of: Title of the study, title and date of publication of the systematic review where the study has been selected, year of publication of the study, date of retraction, and health area (using Cochrane Library topic classifications).

Commented [JW28]: date of retraction would be better than year. Date of publication of Cochrane review would be good too

2.4. Sample Size

The sample size is primarily determined by feasibility, as the application of the INSPECT-SR tool is time-intensive (estimated to take a median time of 45 minutes per trial)[3] and requires a detailed assessment of each trial. We anticipate including 50 trials.

Commented [JW29]: can cite the preprint for I-SR here

Commented [JW30]: Why approximate here? It is under our control, we should select a number in advance (otherwise, how will anyone know that we didn't just stop when we liked our answer?)

To evaluate the precision achievable with this sample size, we considered the estimation of proportions. Under the assumption that a proportion of 70% of the retracted trials will end up with serious INSPECT SR concerns, a sample of 50 trials yields a standard error of approximately 0.065, corresponding to a 95% confidence interval width of ±13%.

This level of precision can be acceptable for an exploratory methodological study, the aim of which is to identify patterns of agreement and potential discrepancies between assessment tools, rather than to produce highly precise estimates[11].

2.5. Analysis Plan

The primary data for analysis will consist of domain-level and overall judgements from the INSPECT-SR assessments and domain-level judgements from the RoB1 assessments. We will use descriptive statistics to summarise the characteristics of the studies and describe the distribution of judgements for both tools. Additionally, we will summarise the number of RoB1 domains at high risk for trials with no, some, or serious INSPECT SR concerns. Moreover, our key metric, which is the proportion of trials with serious INSPECT SR concerns, will be reported with a 95% CI.

Finally, cross-tabulations will be constructed to examine the overlap between trial judgments using INSPECT-SR and RoB1.

For each domain d in the list of the RoB1 domains, i.e. in (Random sequence generation, Allocation concealment, Blinding of participants and personnel, Blinding of outcome assessment, Incomplete outcome data, Selective reporting, Other bias), we will have the plot:

	INSPECT-SR Overall Judgement		
Domain d	No Concerns	Some Concerns	Serious Concerns
Low Risk of Bias			
Some Risk of Bias			
High Risk of Bias			

Moreover, the following plot will summarise the number of domains with a high risk of bias. We will also consider, as a sensitive analysis, the number of domains with a high risk of bias, excluding the domain **Other Bias,** because it is inconsistently used by reviewers[12]

	INSPECT-SR Overall Judgement		
	No Concerns	Some Concerns	Serious Concerns
Number of Domains with High Risk of Bias	**Median (min, max)**	**Median (min, max)**	**Median (min, max)**
Number of Domains with High Risk of Bias Excluding (Other bias)	**Median (min, max)**	**Median (min, max)**	**Median (min, max)**

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